Technische Hochschule Ingolstadt

Fakultät Maschinenbau

Autonomes Fliegen

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Inhalt der Vorlesung 1/2

Kapitel

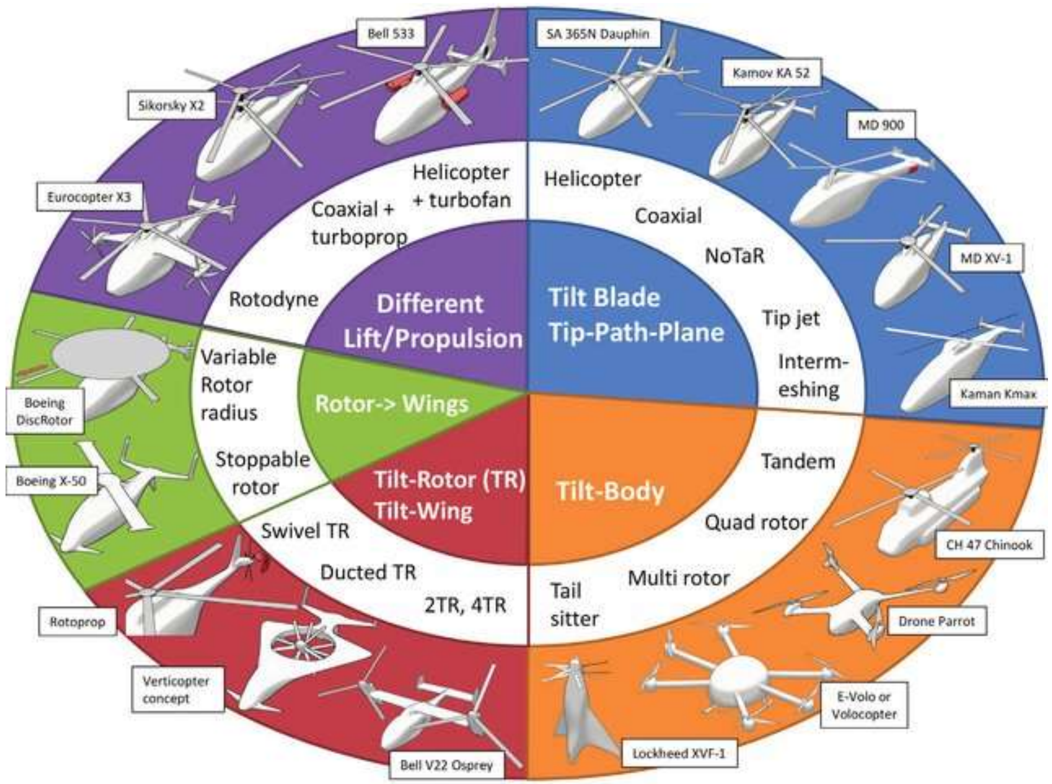
- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (2 W)**
- **Kapitel 4: Flight State Controller (PD Cascade) (1W)**
- **Kapitel 4: Perception und Sensing (1 W)**
- **Kapitel 5: Pfadplanung (4 W)**
- **Kapitel 7: Navigation (2 W)**
- **Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)**

References

- D. Mellinger, "Trajectory Generation and Control for Quadrotors" Ph.D. thesis, University of Pennsylvania, Philadelphia, USA, 2012.
- Modelling and Control of Mini-Flying Machines, 2005.
- D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, 2011, pp. 2520-2525, doi: 10.1109/ICRA.2011.5980409.
- D. McLean, Automatic Flight Control Systems, Prentice Hall International (UK) Ltd, Cambridge, Great Britain, 1990.

Modellierung und Simulation von UAVs

Overview of the different types of rotorcraft concepts



Basset, P. M. et al. "Rotary Wing UAV pre-sizing : Past and Present Methodological Approaches at Onera." (2014).

- **Preliminary design**

Configuration 1 Configuration 2 Configuration 3 Configuration 4 Configuration 5

Basset, P. M. et al. “Rotary Wing UAV pre-sizing : Past and Present Methodological Approaches at Onera.” (2014).

Non-linear Model of a quadrotor

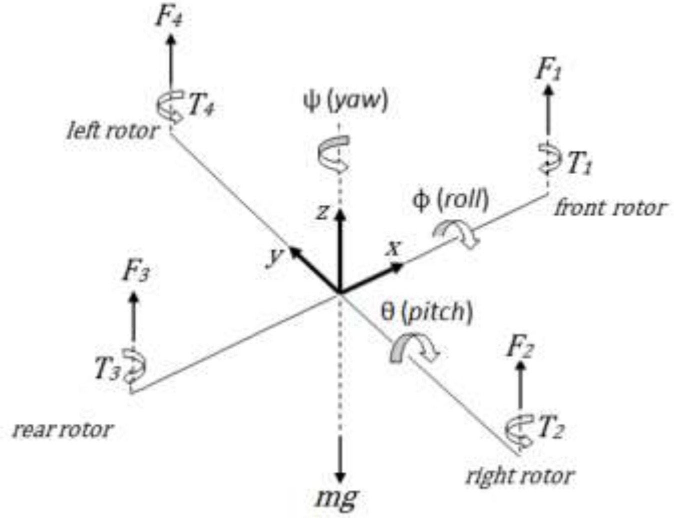
- **Modeling Assumptions**
- **Reference Frames**
- **Kinematics**
- **Dynamics**
- **Control**
- **Simulation**

Assumptions

- **Quadrotor is a rigid body.**
- **Propellers are rigid.**
- **Quadrotor frame is symmetrical.**
- **Mass center and geometric center coincide.**
- **Motor inertia small and neglected.**

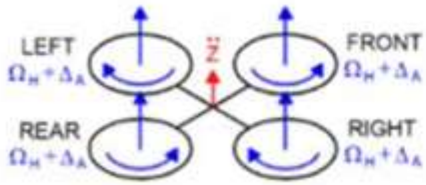
- **Forces and torques generated by each propeller.**

$$\left.\begin{aligned} U_1 &= F_1 + F_2 + F_3 + F_4 \\ U_2 &= F_4 - F_2 \\ U_3 &= F_3 - F_1 \\ U_4 &= T_2 + T_4 - T_1 - T_3 \end{aligned}\right\}$$



- **Four control inputs are used to control quadrotor.**

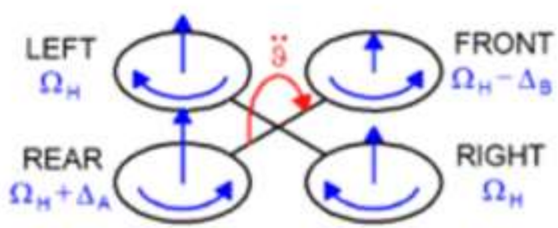
- **Thrust.**


LEFT
$\Omega_H + \Delta_A$
FRONT
$\Omega_H + \Delta_A$
REAR
$\Omega_H + \Delta_A$
RIGHT
$\Omega_H + \Delta_A$
$\ddot{z}$

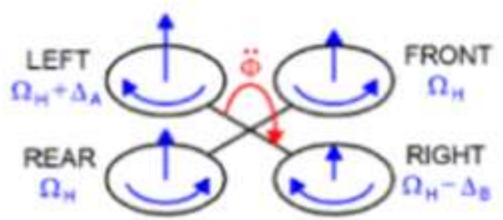
Modellierung und Simulation von UAVs

Functionality 2

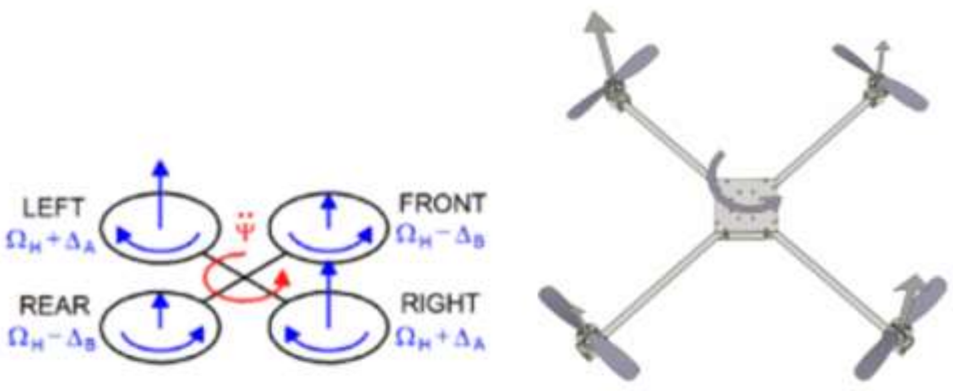
- **Pitch.**



- **Roll.**


LEFT
$\Omega_H + \Delta_A$
FRONT
Ω_H
$\ddot{\Phi}$
REAR
Ω_H
RIGHT
$\Omega_H - \Delta_B$

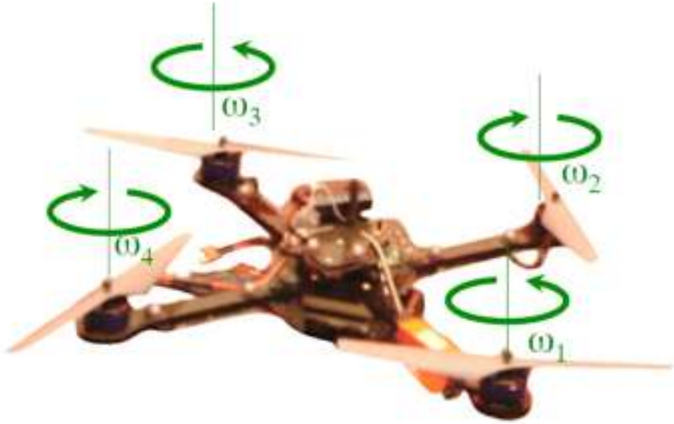
- **Yaw.**



- **Necessity of different frames:**
  - Actuator, inputs and forces act on the body (Body frame).
  - IMU, accelerometers measures quantities in the body frame.
  - GPS measures position in the inertial frame (Inertial frame).
- **Proposed dynamical system model is carried out in the inertial frame.**

Quadrotor

- An UAV with four rotors.
- 6 DOF - 4 inputs $= 2 \neq 0 \Rightarrow$ underactuated
- Adjascent rotors have opposite sense of rotation to balance the total angular momentum.
- Can perform VTOL, hover and slow precise movements



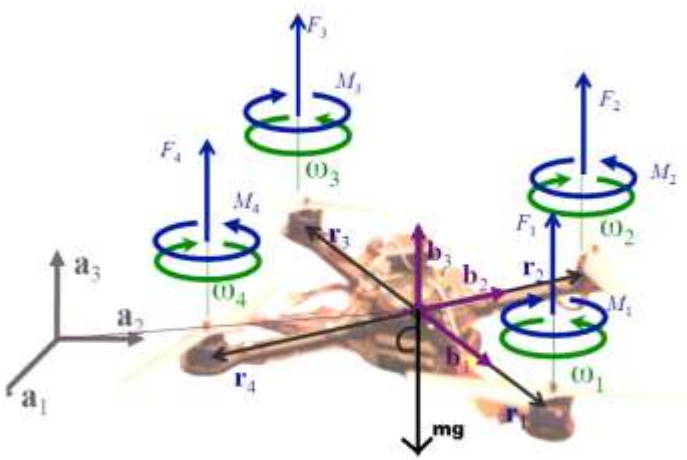
Components of Autonomous Flight

- **State Estimation:** Estimate the linear and angular positions and velocities.
  - Motion Capture System
  - Global Positioning System(GPS)
  - Simulataneous Localization and Mapping(SLAM)
- **Control:** Command actuators to produce desired motion
- **Mapping:** Vehicle must be able to know its surroundings
- **Planning:** The vehicle must be able to compute a trajectory given a set of obstacle and a destination

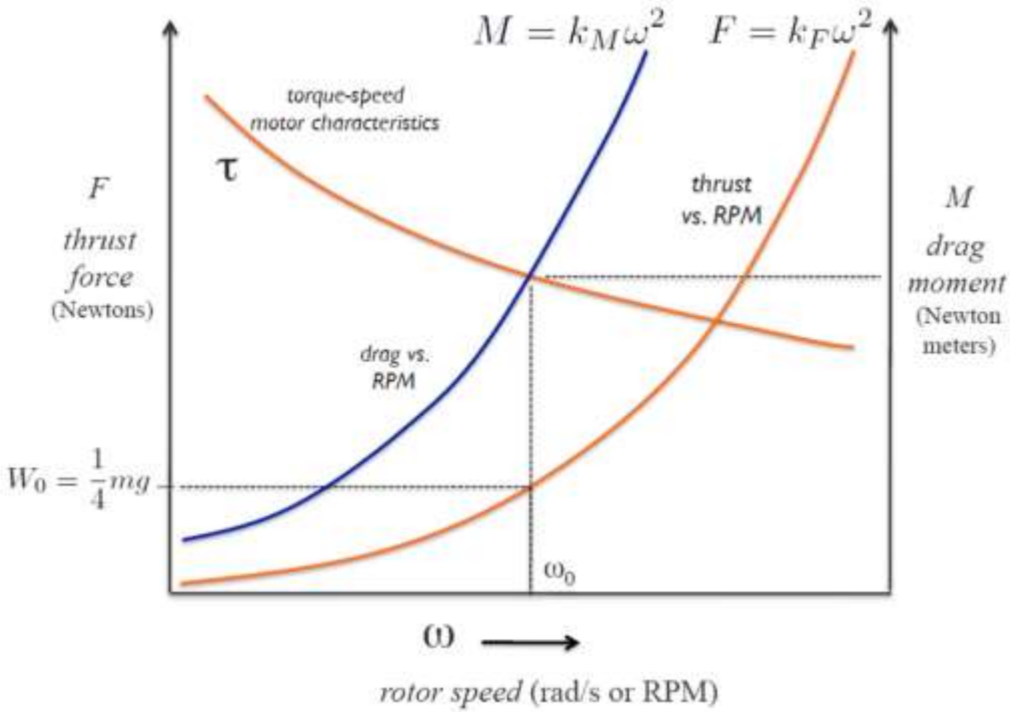
Challenges

- **Underactuated mechanical system**
- **Weight of components**
- **Lack of reliable power source**
- **Legal aspects of use**

- The forces acting on the system are the thrusts F_i from each of the rotor and the force of gravity $-mg\mathbf{a}_3$.
- The moments acting on the system are the moments due to each of the thrust and the drag moment M_i which is generated due to the propellor rotation.



Thrust F_i, Drag Moment M_i and Motor Torque Speed Chacteristics τ vs ω



- Motor Speeds at Hover Configuration:

$$k_F \omega_0^2 = \frac{mg}{4} \tag{1}$$

- Motor Torques and Drag Torque (They have same magnitude but opposite signs):

$$M_i = \tau_i = k_M \omega_i^2 \tag{2}$$

- Thrust

$$F_i = k_F \omega_i^2 \tag{3}$$

- Resultant Force:

$$F = F_1 + F_2 + F_3 + F_4 - mg\mathbf{a}_3 \tag{4}$$

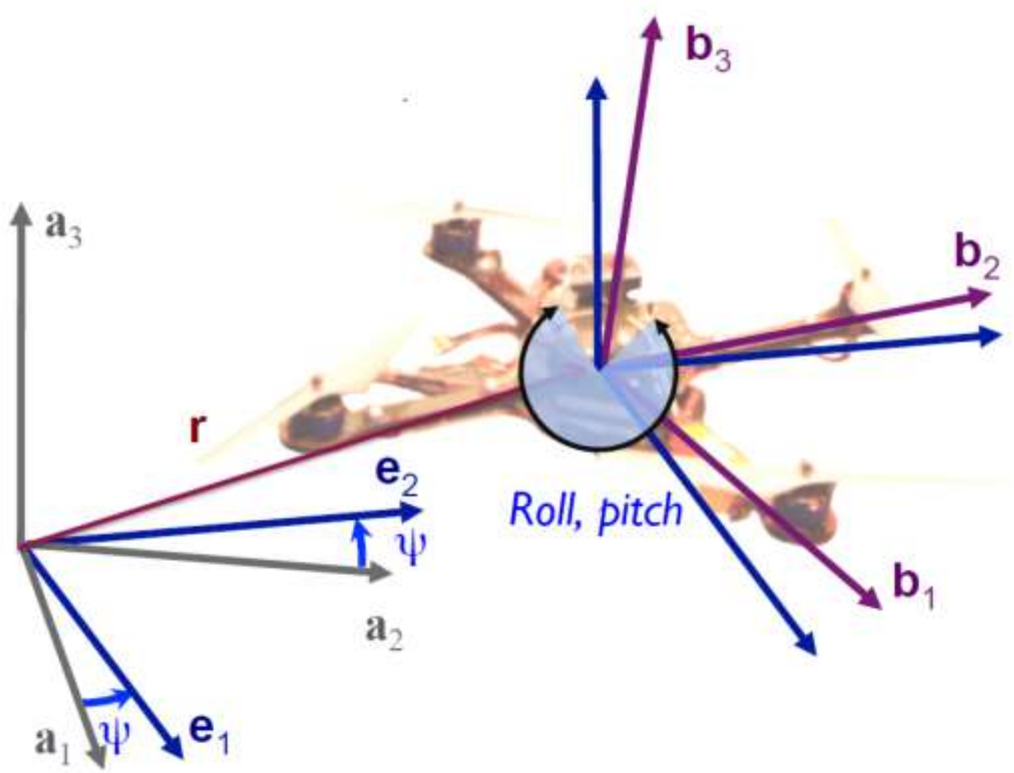
- Resultant Moment:

$$M = r_1 \times F_1 + r_2 \times F_2 + r_3 \times F_3 + r_4 \times F_4 + M_1 + M_2 + M_3 + M_4 \tag{5}$$

In Equilibrium, the resulant force and torque are zero. If it is non zero then the robot accelerates. A combination of motor thrusts and the weight determines the net linear and angular acceleration of the robot.

Modellierung und Simulation von UAVs

Quadrotor Dynamics



- Rotation Matrix: Following the ZXY Euler Angles

$$ {}^{W}R_B = \begin{bmatrix} c\psi c\theta - s\phi s\psi s\theta & -c\phi s\psi & c\psi s\theta + c\theta s\phi s\psi \\ c\theta s\psi + s\phi c\psi s\theta & c\phi c\psi & s\psi s\theta - c\theta s\phi c\psi \\ -c\phi s\theta & s\phi & c\phi c\theta \end{bmatrix} \quad (6) $$

$\phi =$ Roll, $\theta =$ Pitch and $\psi =$Yaw

- Linear Motion Equation in World Frame $\mathbf{a}_1\mathbf{a}_2\mathbf{a}_3$

$$m\ddot{r} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + {}^W R_B \begin{bmatrix} 0 \\ 0 \\ F_1 + F_2 + F_3 + F_4 \end{bmatrix} \qquad (7)$$

- Angular Motion Equation in the Body Frame $\mathbf{b}_1\mathbf{b}_2\mathbf{b}_3$

$$\mathbf{I} \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} L(F_2 - F_4) \\ L(F_3 - F_1) \\ M_2 + M_4 - M_1 - M_3 \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \mathbf{I} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \qquad (8)$$

where, $\mathbf{r} = [x\ y\ z]^T$, I = Moment of Interia, m = Mass of the System, $[p\ q\ r]$: Body Angular Velocities

- The relation ship between body angular velocities and the rate of change of Roll, Pitch and Yaw.

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} c\theta & 0 & -c\phi s\theta \\ 0 & 1 & s\phi \\ s\theta & 0 & c\phi c\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{9}$$

Inputs to the dynamical system model

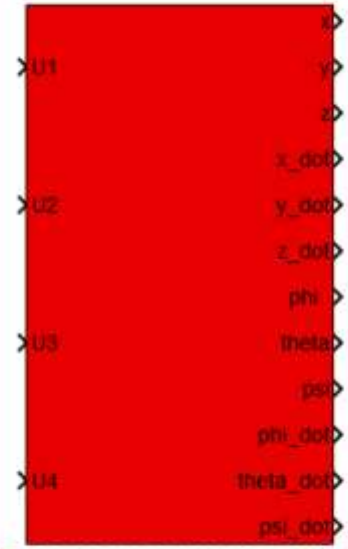
- Thrust Input: $u_1 = F_1 + F_2 + F_3 + F_4$
- Torque Input: $u_2 = \begin{bmatrix} L(F_2 - F_4) \\ L(F_3 - F_1) \\ M_2 + M_4 - M_1 - M_3 \end{bmatrix}$

Summery

$$\begin{bmatrix}\ddot{x}\\ \ddot{y}\\ \ddot{z}\end{bmatrix} = -\begin{bmatrix}0\\0\\g\end{bmatrix} + L_{EB}\begin{bmatrix}0\\0\\U_1/m\end{bmatrix} - (K_t/m)\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{z}\end{bmatrix}$$

$$\begin{bmatrix}\dot{p}\\ \dot{q}\\ \dot{r}\end{bmatrix} = \begin{bmatrix}(I_y - I_z)qr/I_x\\ (I_z - I_x)pr/I_y\\ (I_x - I_y)pq/I_z\end{bmatrix} + \begin{bmatrix}U_2 d/I_x\\ U_3 d/I_y\\ U_4/I_z\end{bmatrix}$$

$$\begin{bmatrix}\dot{\phi}\\ \dot{\theta}\\ \dot{\psi}\end{bmatrix} = \begin{bmatrix}1 & sin(\phi)tan(\theta) & cos(\phi)tan(\theta)\\ 0 & cos(\phi) & -sin(\phi)\\ 0 & sin(\phi)/cos(\theta) & cos(\phi)/cos(\theta)\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix}$$



QUADROTOR NONLINEAR DYNAMICS